

ASHWIN K K

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LinkedIn

Portfolio

SUMMARY

Electronics and Communication Engineer with hands-on knowledge in the areas of embedded systems, IoT, and wireless communication technologies. Proficient in Embedded C, ESP32, Arduino, PCB design, soldering, and electronic circuit testing. Familiar with communication protocols. Seeking opportunities to develop innovative and reliable engineering solutions.

EXPERIENCE

LS Automotive India Pvt ltd, Chennai

June 2026 – Aug 2026

Graduate Engineer Trainee

- Worked in the **Assembly Department**, Supporting production and assembly operations for automotive components.
- Assisted **machine operations and quality checks** according to standard operating procedures.
- Gained understanding of **automotive components, subassemblies and production line workflow**.
- Followed **5S, safety procedures, quality standards**, and workplace protocols.

SKILLS

Programming Languages: Embedded C, SQL

Tools: KiCad, MPLAB, Ltspice, MS Excel, MS Word

Technical Knowledge: UART, SPI, I2C

Soft Skills: Communication, Teamwork, Observation, Creativity

PROJECTS

LoRa-based V2X Communication System

- Developed And tested a V2X communication system prototype using ESP32, Arduino Nano, and LoRa (SX1278).
- Implemented various components such as V2V, V2I, and V2P communication for traffic and safety applications.
- Generated collision warning, traffic-signal, and pedestrian alert notifications.

Blackbox System for Emergency Response

- Developed a vehicle SOS system for accident detection and emergency response, designed to detect accident events.
- Sent emergency alerts with GPS location, date, and timestamp information to provide essential accident details.
- Stored accident-related data in memory and evaluated sensor response accuracy to assess overall system performance.

Vision-Based Intelligent Auditorium Automation

- Developed a vision-based automation system using the YOLO algorithm to optimize electrical energy consumption.
- Automatically controlled lights and fans based on detected human presence.
- Analysed how energy can be effectively utilized and also monitored the number of people inside halls.

EDUCATION

B.Tech. in Electronics and Communication Engineering

2022 - 2026

College of Engineering Thalassery

Diploma in Electronics Engineering

2020 - 2022

Government Polytechnic College Mattanur

CERTIFICATES

Embedded System Design and Programming

Completed a two-week internship program focused on PIC microcontroller programming. Developed practical knowledge of various interfacing techniques and PCB design.

Introduction to IoT and Embedded Systems

Learned the basics of IoT devices, Embedded systems, and communication protocols and understood how connected devices communicate and exchange data between them.

Technical Support Fundamentals

Gained the knowledge of troubleshooting and resolving common IT-related problems.